

Newton on KKT	KKT Analytical
$\min f(x) = x_1^2 + 5x_2^2, \text{ s.t. } g(x) = 4 - x_1 - x_2 \leq 0$	$\min f(x) = x_1^2 + 5x_2^2, \text{ s.t. } g(x) = x_1 + x_2 - 4 \leq 0$
	Slack: $g(x) - s^2 = 0$, so $l(x, \lambda, s) = f(x) - \lambda(g(x) - s^2)$
	$\frac{\partial l}{\partial s} = 2\lambda s = 0 \Rightarrow \lambda = 0$ (inactive) or $s = 0$ (active)
	Test: $X_{\text{unc}} = (0, 0)$; $g(X_{\text{unc}}) = -4 < 0 \quad \therefore$ infeasible $\Rightarrow \lambda \neq 0$ $\xrightarrow{2\lambda s=0} s = 0$, constraint active
$l(x, \lambda) = x_1^2 + 5x_2^2 - \lambda(x_1 + x_2 - 4)$ (same l , implicit in $\nabla_x l$)	$l(x, \lambda) = x_1^2 + 5x_2^2 - \lambda(x_1 + x_2 - 4)$ (set $s = 0$)
Assert $r(x, \lambda) = 0$ and J ; note $J = \begin{bmatrix} H_l & -\nabla_x g \\ \nabla_x g^\top & 0 \end{bmatrix}$:	Same r and J arrived at from $\nabla_x l = 0$ and $g = 0$:
$r = \begin{bmatrix} \nabla_x l \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} \quad J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$	$r = \begin{bmatrix} \nabla_x l \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} \quad J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$
Init: $z_0 = (0, 0, 0)^\top$; evaluate r at z_0 :	(no iterate — constraint known active from test above)
$r(z_0) = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix}_{z_0=(0,0,0)} = \begin{bmatrix} 2(0) - 0 \\ 10(0) - 0 \\ 0 + 0 - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -4 \end{bmatrix}$	
Newton step: $J \Delta z = -r(z_0)$, i.e. $-r(z_0) = (0, 0, 4)^\top$:	$\nabla_x l = H_f x - \lambda \nabla_x g = 0 \Rightarrow x^*(\lambda) = H_f^{-1} \lambda \nabla_x g$
$J \Delta z = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \end{bmatrix} = 0 \Rightarrow x^*(\lambda) = \begin{bmatrix} \lambda/2 \\ \lambda/10 \end{bmatrix}$
Row 1: $2\Delta x_1 - \Delta \lambda = 0 \Rightarrow \Delta x_1 = \frac{\Delta \lambda}{2}$ Row 2: $10\Delta x_2 - \Delta \lambda = 0 \Rightarrow \Delta x_2 = \frac{\Delta \lambda}{10}$	
Row 3: $\frac{\Delta \lambda}{2} + \frac{\Delta \lambda}{10} = 4 \Rightarrow 6\Delta \lambda = 40 \Rightarrow \Delta \lambda = \frac{20}{3}$	Sub $x^*(\lambda)$ into $g = 0$: $\frac{\lambda}{2} + \frac{\lambda}{10} = 4 \Rightarrow 10\lambda + 2\lambda = 80 \Rightarrow \lambda = \frac{20}{3}$
$\Delta x_1 = \frac{10}{3}, \Delta x_2 = \frac{2}{3}; z_1 = z_0 + \Delta z = (\frac{10}{3}, \frac{2}{3}, \frac{20}{3})$	$x^* = \left(\frac{10}{3}, \frac{2}{3}\right)^\top$
$f(x^*) = \left(\frac{10}{3}\right)^2 + 5\left(\frac{2}{3}\right)^2 = \frac{100}{9} + \frac{20}{9} = \frac{40}{3}$	$f(x^*) = \left(\frac{10}{3}\right)^2 + 5\left(\frac{2}{3}\right)^2 = \frac{100}{9} + \frac{20}{9} = \frac{40}{3}$

Newton’s method applied to KKT (same example: $\min x_1^2 + 5x_2^2$ s.t. $x_1 + x_2 = 4$, starting at $z_0 = (0, 0, 0)$)

Step 1 — residual at z_0 : **Step 3 — solve $J \Delta z = -r(z_0)$:**

$$r(z_0) = \begin{bmatrix} 2(0) - 0 \\ 10(0) - 0 \\ 0 + 0 - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$$

Step 2 — Jacobian (constant here since r is linear):

$$J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

Row 1: $2\Delta x_1 - \Delta \lambda = 0 \Rightarrow \Delta x_1 = \frac{\Delta \lambda}{2}$
Row 2: $10\Delta x_2 - \Delta \lambda = 0 \Rightarrow \Delta x_2 = \frac{\Delta \lambda}{10}$
Row 3: $\Delta x_1 + \Delta x_2 = 4 \Rightarrow \frac{\Delta \lambda}{2} + \frac{\Delta \lambda}{10} = 4 \Rightarrow 6\Delta \lambda = 40 \Rightarrow \Delta \lambda = \frac{20}{3}$

$$\Delta x_1 = \frac{10}{3} \qquad \Delta x_2 = \frac{2}{3}$$

Step 4 — update: $z_1 = z_0 + \Delta z$

$$z_1 = \left(\frac{10}{3}, \frac{2}{3}, \frac{20}{3}\right) = x^* \quad \checkmark \quad (\text{one step: KKT system is linear})$$

Newton — unconstrained $p_0 = -H_f(X_0)^{-1} \nabla f(X_0)$
 $f(x_1, x_2) = x_1^2 + 5x_2^2, \quad X_0 = (3, 1)^\top$

$$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix} \Rightarrow \nabla f(X_0) = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

$$H_f = \begin{bmatrix} 2 & 0 \\ 0 & 10 \end{bmatrix} \Rightarrow H_f^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix}$$

$$p_0 = - \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix} \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$$

Line search: $\hat{X}_1 = \begin{bmatrix} 3 - 3\alpha \\ 1 - \alpha \end{bmatrix}$

$$\phi(\alpha) = (3 - 3\alpha)^2 + 5(1 - \alpha)^2 = 14 - 28\alpha + 14\alpha^2$$

$$\frac{d\phi}{d\alpha} = -28 + 28\alpha = 0 \Rightarrow \alpha_0 = 1$$

Update: $X_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + 1 \cdot \begin{bmatrix} -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$f(X_1) = 0$ — one step, exact (quadratic f , $\alpha = 1$ always)

BFGS quasi-Newton $p_0 = -\tilde{\nabla}^2 f(X_0)^{-1} \nabla f(X_0), \quad B_0 = I$
 $f(x_1, x_2) = x_1^2 + 5x_2^2, \quad X_0 = (3, 1)^\top$

$$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix} \Rightarrow \nabla f(X_0) = \begin{bmatrix} 2 \cdot 3 \\ 10 \cdot 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

Candidate: $\hat{X}_1 = X_0 + \alpha p_0 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \alpha \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 3 - 6\alpha \\ 1 - 10\alpha \end{bmatrix}$

$$\begin{aligned} \phi(\alpha) &= (3 - 6\alpha)^2 + 5(1 - 10\alpha)^2 \\ &= 9 - 36\alpha + 36\alpha^2 + 5(1 - 20\alpha + 100\alpha^2) \\ &= 14 - 136\alpha + 536\alpha^2 \end{aligned}$$

$$\frac{d\phi}{d\alpha} = -136 + 1072\alpha = 0 \Rightarrow \alpha_0 = \frac{136}{1072} \approx 0.1269$$

$$X_1 = X_0 + \alpha_0 \nabla f(X_0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - 0.1269 \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 2.239 \\ -0.269 \end{bmatrix}$$

BFGS vectors:

$$s_0 = -\alpha_0 \nabla f(X_0) = -0.1269 \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -0.7614 \\ -1.269 \end{bmatrix}$$

$$y_0 = \nabla f(X_1) - \nabla f(X_0) = \begin{bmatrix} 2 \cdot 2.239 \\ 10 \cdot (-0.269) \end{bmatrix} - \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -1.522 \\ -12.689 \end{bmatrix}$$

Plus Quest

$$p_0 = -\hat{V} \nabla f(x_0)$$

$$f(x_1, x_2) = x_1^2 + 5x_2^2 \quad x \in \mathbb{R}^2 \quad x_0 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}^T$$

$$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix} \quad \nabla f(x_0) = \begin{bmatrix} 2 \cdot 3 \\ 10 \cdot 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

$$x_1 = x_0 + \alpha p_0 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \alpha \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 3 - 6\alpha \\ 1 - 10\alpha \end{bmatrix}$$

$$\phi(\alpha) = (3 - 6\alpha)^2 + 5(1 - 10\alpha)^2$$

$$= 9 - 36\alpha + 36\alpha^2 + 5(1 - 20\alpha + 100\alpha^2)$$

$$= 9 - 36\alpha + 36\alpha^2 + 5 - 100\alpha + 500\alpha^2$$

$$= 14 - 136\alpha + 536\alpha^2$$

$$\phi(\alpha)$$

$$\frac{d\phi}{d\alpha} = -136 + 1072\alpha \rightarrow \alpha_0 = \frac{136}{1072} = 0.1269$$

$$x_1 = x_0 - \alpha \nabla f(x_0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - 0.1269 \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 2.239 \\ -0.269 \end{bmatrix}$$

$$s_0 = -\alpha_0 \nabla f(x_0) = -0.1269 \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -0.7614 \\ -1.269 \end{bmatrix}$$

$$y_0 = \nabla f(x_1) - \nabla f(x_0)$$

$$y_0 = \begin{bmatrix} 2 \cdot 2.239 \\ 10 \cdot (-0.269) \end{bmatrix} - \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -1.522 \\ -12.689 \end{bmatrix}$$